

Question No. 01

We can roughly divide SONET networks into _____ categories:

- A. One
- B. Two
- C. Three
- D. Four

SONET Networks

We can roughly divide SONET networks into three categories:

1. Linear Networks
2. Ring Networks
3. Mesh Networks

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Question No. 02

Hub device operates on the _____ Layer.

- A. Transport Layer
- B. Network Layer
- C. Data Link Layer
- D. Physical Layer

OSI - Model

Application Layer	7
Presentation Layer	6
Session Layer	5
Transport Layer	4
Network Layer	3
Data Link Layer	2
Physical Layer	1

☐ Router (1, 2, 3)

☐ Switch (1, 2)

☐ HUB

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Question No. 03

Architecture of a SONET system consists of _____, _____ and _____.

- A. Signals
- B. Devices
- C. Connections
- D. All of the above

SONET - Synchronous Optical Network

Architecture of a SONET system consists of signals, devices, and connections

1. Signals

- Synchronous Transport Signals (STS)
- Optical Carriers (OCs)
- Synchronous Transport Module (STM)

2. Devices

- STS Mux/Demux
- Regenerators
- Add-Drop Multiplexer and Terminals

3. Connections

- Section
- Line
- Path

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Question No. 04

FTTB stands for _____

- A. Fiber To The Block
- B. Fiber To The Building**
- C. Fiber To The Box
- D. All of the above

- 1. FTTH - Fiber To The Home
- 2. FTTC - Fiber To The Curb
- 3. FTTB - Fiber To The Building
- 4. FTTN - Fiber To The Network
- 5. NT - Network Termination
- 6. ONU - Optical Network Unit
- 7. POTS – Plain Old Telephone Service
- 8. SDV - Switched Digital Video
- 9. LAN - Local Area Network
- 10. WAN - Wide Area Network
- 11. MAN - Metropolitan Area Network
- 12. PAN - Personal Area Network
- 13. BSS - Basic Service Set
- 14. ESS - Extended Service Set
- 15. DCF - Distributed Coordination Function
- 16. PCF - Point Coordination Function



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Question No. 05

The T4 has _____ wires.

- A. 1 Wires
- B. 2 Wires
- C. 4 Wires
- D. None of the above

Fast Ethernet implementations

TX → 2 Wires

FX → 2 Wires

T4 → 4 Wires

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Question No. 06

Which of the following include in the types of stations?

- A. No-Transition Mobility
- B. BSS-Transition Mobility
- C. ESS-Transition Mobility
- D. All of the above

Types of Stations

There are three type of stations:

1. No-Transition Mobility
2. BSS-Transition Mobility
3. ESS-Transition Mobility

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Question No. 07

Each Signal is composed of _____ frames

- A. 6000
- B. 7000
- C. 8000
- D. 9000

SONET Frames

STS - Synchronous Transport Signal

ST-N Each Signal is composed of 8000 frames
Each frame is a two-dimensional matrix of bytes with 9 rows by $90 \times n$ columns

ST-1 Frame is 9 rows 90×1 columns (810 bytes)

ST-3 Frame is 9 rows 90×3 columns (2430 bytes)

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Question No. 08

Ethernet LAN was Developed in:

- A. 1960s
- B. 1970s
- C. 1980s
- D. 1990ss

Ethernet Evolution

Ethernet Evolution

The Ethernet LAN was developed in the 1970s

Since then, it has gone through four generations:

Standard Ethernet	(10 Mbps)
Fast Ethernet	(100 Mbps)
Gigabit Ethernet	(1 Gbps)
10 Gigabit Ethernet	(10 Gbps)

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 09

A frame has _____.

- A. Fixed Size
- B. Variable Size
- C. Both A and B
- D. None of the above

Frame Size

Frames can be of:

Fixed Size

Size acts as a boundary/delimiter

Variable Size

How to define Beginning and End of a Frame?

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Question No. 10

Which of the following is the Control Access Method?

- A. Reservation
- B. Polling
- C. Token Passing
- D. All of the above

Control Access Methods

We discuss three controlled-access methods:

1. Reservation
2. Polling
3. Token Passing



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Question No. 11

Which of the following include in the connecting device?

- A. Hubs
- B. Switches
- C. Routers
- D. All of the above

Three kinds of Connecting Devices

- 1. Hubs
- 2. Switches
- 3. Routers

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Question No. 12

Bluetooth defines two types of networks:

- A. Piconet
- B. Scatternet
- C. Both A and B
- D. None of the above

Bluetooth

Bluetooth defines two types of networks:

- 1. Piconet
- 2. Scatternet

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 13

The 1100 subtype represent the meaning:

- A. RTS - Request to Send
- B. CTS - Clear to Send
- C. ACK - Acknowledgement
- D. None of the above

Values of Subfields in Control Frames

Subtype	Meaning
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 14

Which of the following is NOT include in SONET networks categories?

- A. Linear Networks
- B. Ring Networks
- C. Bus Network
- D. Mesh Networks

SONET Networks

We can roughly divide SONET networks into three categories:

- 1. Linear Networks
- 2. Ring Networks
- 3. Mesh Networks

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Question No. 15

ATM stand for _____.

- A. Asynchronous Transfer Mode
- B. Automatic Teller Machine
- C. Both A and B
- D. None of the above

ATM

ATM - Asynchronous Transfer Mode



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Question No. 16

The telephone network has following office
_____.

- A. End offices
- B. Tandem offices
- C. Regional offices
- D. All of the above

Telephone

The telephone network has several levels of switching offices:

1. End offices
2. Tandem offices
3. Regional offices

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Question No. 17

The Thin Network has _____.

- A. 10 Base 5
- B. 10 Base 2
- C. 10 Base T
- D. 10 Base F

Network

Thick Network → 10 Base 5

Thin Network → 10 Base 2

Twisted Network → 10 Base T

Fiber Network → 10 Base F

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Question No. 18

The Time is divided into _____.

- A. Frame
- B. Packets
- C. Interval
- D. None of the above

Intervals

Time is divided into intervals

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Question No. 19

The primary device controls the _____; the secondary devices follow its _____.

- A. Links
- B. Instruction
- C. Both A and B
- D. None of the above

Polling

Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

The primary device controls the link; the secondary devices follow its instructions

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Question No. 20

Which of the following protocol include in Channelization:

- A. FDMA
- B. TDMA
- C. CDMA
- D. All of the above

Channelization

Channelization has three protocols:

1. Frequency Division Multiple Access (FDMA)
2. Time Division Multiple Access (TDMA)
3. Code Division Multiple Access (CDMA)

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 21

Switch device operates on the _____ Layer.

- A. Transport Layer
- B. Network Layer
- C. Data Link Layer
- D. Physical Layer

OSI - Model

Application Layer	7	
Presentation Layer	6	
Session Layer	5	
Transport Layer	4	
Network Layer	3	☐ Router (1, 2, 3)
Data Link Layer	2	☐ Switch (1, 2)
Physical Layer	1	☐ HUB

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 22

Which of the following is the benefits / advantages of VLANs?

- A. Save Time
- B. Save Cost
- C. Security
- D. All of the above

Advantages of VLANs

1. Cost and Time Reduction
2. Creating virtual Workgroups
3. Security

Configuration of a VLAN

1. Manually
2. Semi-Automatically
3. Automatically

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 23

Which of the following is NOT include in Connections of Synchronous Optical Network?

- A. Section
- B. Line
- C. Transfer
- D. Path

SONET - Synchronous Optical Network

Architecture of a SONET system consists of signals, devices, and connections

1. Signals

- Synchronous Transport Signals (STS)
- Optical Carriers (OCs)
- Synchronous Transport Module (STM)

2. Devices

- STS Mux/Demux
- Regenerators
- Add-Drop Multiplexer and Terminals

3. Connections

- Section
- Line
- Path

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 24

If 8000 frames transfer from sources to destination than find out no of bytes at STS-1 signal?

- A. 6,480 bytes
- B. 6,480,000 bytes**
- C. 648,000 bytes
- D. 648 bytes

SONET Frames

STS - Synchronous Transport Signal

STS-N Each Signal is composed of 8000 frames
Each frame is a two-dimensional matrix of bytes with 9 rows by $90 \times n$ columns

STS-1 Frame is 9 rows 90×1 columns (810 bytes)

STS-3 Frame is 9 rows 90×3 columns (2430 bytes)

1 Kilobyte = 1000 bytes

1 Megabit/s = 125 Kilobytes/sec

STS-1 = Frame $\times$ 9 Rows ($90 \times N$)

= 8000×9 (90×1)

= 8000×9 (90)

= 8000×810

= 6,480,000 bytes

= $(6,480,000 / 1000) / 125$

= $6,480 / 125$

= **51.84**

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Question No. 25

The Gigabit Ethernet hardware operates at a rate of _____.

- A. 10 Mbps
- B. 100 Mbps
- C. 1000 Mbps
- D. None of the above

Rates

1 Kilobyte = 1000 bytes

1 Megabit/s = 125 Kilobytes/sec

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Question No. 26

DSL stands for _____.

- A. Distance Subscriber Line
- B. Digital Subscriber Line**
- C. Distance Subscriber Location
- D. None of the above

- 1. CM - Cable Modem
- 2. CMTS - Cable Modem Transmission System
- 3. HFC - Hybrid Fiber Coaxial
- 4. CATV - Community Antenna Television
- 5. DSL - Digital Subscriber Line
- 6. LATA - Local-Access Transport Areas
- 7. LEC - Local Exchange Carrier
- 8. IXC - Interexchange Carriers
- 9. POP - Points of Presence
- 10. MTP - Message Transfer Part
- 11. SCCP - Signaling Connection Control Point
- 12. TCAP - Transaction Capabilities Application Port
- 13. TUP - Telephone User Port
- 14. LLC - Logical Link Control
- 15. MAC - Media Access Control

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Question No. 27

Telephone companies provide two types of services:

- A. Analog Services
- B. Digital Services
- C. Both A and B
- D. None of the above

Telephone Services

Telephone companies provide two types of services:

1. Analog Services
2. Digital Services

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Question No. 28

The Thick Network has _____.

- A. 10 Base 5
- B. 10 Base 2
- C. 10 Base T
- D. 10 Base F

Network

Thick Network → 10 Base 5

Thin Network → 10 Base 2

Twisted Network → 10 Base T

Fiber Network → 10 Base F

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Question No. 29

In CDMA the Bit 1 represent _____.

- A. 0
- B. 1
- C. -1
- D. +1

Data Representation in CDMA

Data Bit 0	→	-1
Data Bit 1	→	+1
Silence	→	0

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Question No. 30

Variable Framing Techniques includes _____.

- A. Bit Oriented Framing
- B. Character Oriented Framing
- C. Both A and B
- D. None of the above

Variable Framing Techniques

Bit Oriented Framing
Character Oriented Framing

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Question No. 31

_____ is a graph in which there is no loop.

- A. Algorithm
- B. **Spanning Tree Algorithm**
- C. Routers
- D. None of the above

Spanning Tree Algorithm

In graph theory, Spanning Tree is a graph in which there is no loop

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Question No. 32

The 1101 subtype represent the meaning:

- A. RTS - Request to Send
- B. CTS - Clear to Send
- C. ACK - Acknowledgement
- D. None of the above

Values of Subfields in Control Frames

Subtype	Meaning
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

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Question No. 33

Which of the following include in the types of frames?

- A. Management Frames
- B. Control Frames
- C. Data Frames
- D. All of the above

Types of Frames

There are three type of frames:

1. Management Frames
2. Control Frames
3. Data Frames

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Question No. 34

_____ protocol uses both flow and error control.

- A. Simple Protocol
- B. Stop-and-Wait Protocol**
- C. Selective-Repeat Protocol
- D. All of the above

DATA-LINK LAYER PROTOCOLS

DATA-LINK LAYER PROTOCOLS ARE

Simple Protocol

Simple protocol has neither flow nor error control

Stop-and-Wait Protocol

Stop-and-Wait protocol uses both flow and error control

Go-Back-N Protocol

Selective-Repeat Protocol

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Question No. 35

How many layers include in SONET standard:

- A. 4
- B. 3
- C. 2
- D. 1

SONET Standard

The SONET standard includes four functions.

1. The Path Layer
2. The Line Layer
3. The Section Layer
4. The Photonic Layer



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Question No. 36

The Fast Ethernet provide the speed of _____

- A. 10 Mbps
- B. 100 Mbps**
- C. 1 Gbps
- D. 10 Gbps

Ethernet Evolution

Ethernet Evolution

The Ethernet LAN was developed in the 1970s



Since then, it has gone through four generations:

Standard Ethernet	(10 Mbps)
Fast Ethernet	(100 Mbps)
Gigabit Ethernet	(1 Gbps)
10 Gigabit Ethernet	(10 Gbps)

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 37

The Fiber Network has _____.

- A. 10 Base 5
- B. 10 Base 2
- C. 10 Base T
- D. 10 Base F

Network

Thick Network → 10 Base 5

Thin Network → 10 Base 2

Twisted Network → 10 Base T

Fiber Network → 10 Base F

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Question No. 38

How many services provided by DLC:

- A. 4
- B. 3**
- C. 2
- D. 1

DLC Services

Following are include in DLC Services:

- Framing**
- Flow Control**
- Error Control**

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Question No. 39

TUP stands for _____.

- A. Transaction User Port
- B. Telephone User Port**
- C. Transfer User Port
- D. None of the above

- 1. CM - Cable Modem
- 2. CMTS - Cable Modem Transmission System
- 3. HFC - Hybrid Fiber Coaxial
- 4. CATV - Community Antenna Television
- 5. DSL - Digital Subscriber Line
- 6. LATA - Local-Access Transport Areas
- 7. LEC - Local Exchange Carrier
- 8. IXC - Interexchange Carriers
- 9. POP - Points of Presence
- 10. MTP - Message Transfer Part
- 11. SCCP - Signaling Connection Control Point
- 12. TCAP - Transaction Capabilities Application Port
- 13. TUP- Telephone User Port
- 14. LLC - Logical Link Control
- 15. MAC - Media Access Control

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Question No. 40

In _____ Both Simple and Stop-and-wait protocols are designed for unidirectional communication.

- A. Packeting
- B. Framing
- C. Piggybacking
- D. Error Control

Piggybacking

In Piggybacking Both Simple and Stop-and-wait protocols are designed for unidirectional communication

MOHSIN RAZA

MSBA, MBA, MIT, M.ed, B.ed, LL.B

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Question No. 41

Router device operates on the _____ Layer.

- A. Transport Layer
- B. Network Layer**
- C. Data Link Layer
- D. Physical Layer

OSI - Model

Application Layer	7
Presentation Layer	6
Session Layer	5
Transport Layer	4
Network Layer	3
Data Link Layer	2
Physical Layer	1

☐ Router (1, 2, 3)

☐ Switch (1, 2)

☐ HUB



SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 42

Which of the following method is used for the configuration of VLANs?

- A. Manually
- B. Semi Automatically
- C. Automatically
- D. All of the above

Advantages of VLANs

1. Cost and Time Reduction
2. Creating virtual Workgroups
3. Security

Configuration of a VLAN

1. Manually
2. Semi-Automatically
3. Automatically

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Question No. 43

Which of the following bit or subtypes is used for RTS?

- A. 1 0 1 0
- B. 1 1 0 0
- C. 1 0 1 1
- D. 1 1 1 1

Values of Subfields in Control Frames

Subtype	Meaning
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 44

Which of the following is NOT include in Signals of Synchronous Optical Network?

- A. STS
- B. RTS**
- C. OCS
- D. STM

SONET - Synchronous Optical Network

Architecture of a SONET system consists of signals, devices, and connections

1. Signals

- Synchronous Transport Signals (STS)
- Optical Carriers (OCs)
- Synchronous Transport Module (STM)

2. Devices

- STS Mux/Demux
- Regenerators
- Add-Drop Multiplexer and Terminals

3. Connections

- Section
- Line
- Path

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 45

If 8000 frames transfer from sources to destination than find out Data Rate of STS-3 signal?

- A. 15.52
- B. 1.52
- C. 155.52
- D. 1555.52

SONET Frames

STS - Synchronous Transport Signal

STS-N Each Signal is composed of 8000 frames
Each frame is a two-dimensional matrix of bytes with 9 rows by $90 \times n$ columns

STS-1 Frame is 9 rows 90×1 columns (810 bytes)

STS-3 Frame is 9 rows 90×3 columns (2430 bytes)

1 Kilobyte = 1000 bytes

1 Megabit/s = 125 Kilobytes/sec

STS-1 = Frame $\times$ 9 Rows ($90 \times N$)

= 8000×9 (90×3)

= 8000×9 (270)

= 8000×810

= 19,440,000 bytes

= $(19,440,000 / 1000) / 125$

= $19,440 / 125$

= 155.52

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 46

The 10 Gigabit Ethernet provide the speed of _____.

- A. 10 Mbps
- B. 100 Mbps
- C. 1 Gbps
- D. 10 Gbps

Ethernet Evolution

Ethernet Evolution

The Ethernet LAN was developed in the 1970s

Since then, it has gone through four generations:

Standard Ethernet	(10 Mbps)
Fast Ethernet	(100 Mbps)
Gigabit Ethernet	(1 Gbps)
10 Gigabit Ethernet	(10 Gbps)

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 47

The TF has _____ wires.

- A. 1 Wires
- B. 2 Wires**
- C. 4 Wires
- D. None of the above

Fast Ethernet implementations

TX → 2 Wires

FX → 2 Wires

T4 → 4 Wires

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 48

In CDMA the Bit 0 represent _____.

- A. 0
- B. 1
- C. -1
- D. +1

Data Representation in CDMA

Data Bit 0	→	-1
Data Bit 1	→	+1
Silence	→	0

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 49

Which of the following is NOT include in DLC Services?

- A. **Packeting**
- B. Framing
- C. Flow Control
- D. Error Control

DLC Services

Following are include in DLC Services:

- Framing**
- Flow Control**
- Error Control**

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**

Question No. 50

How many services provided by DLC:

- A. Simple Protocol
- B. Stop-and-Wait Protocol
- C. Selective-Repeat Protocol
- D. All of the above

DATA-LINK LAYER PROTOCOLS

DATA-LINK LAYER PROTOCOLS ARE

Simple Protocol

Simple protocol has neither flow nor error control

Stop-and-Wait Protocol

Stop-and-Wait protocol uses both flow and error control

Go-Back-N Protocol

Selective-Repeat Protocol

SUBSCRIBE TO **KEY TO SUCCESS** FOR **GUARANTEED SUCCESS**